

Coding Assignment Demo System

DAVID COOPER, Virigina Tech, USA

Understanding programming assignments can be challenging for students, particularly when problem statements are complex or unfamiliar. In this paper, we look at an approach to enhance understanding by embedding interactive components within assignment descriptions. The created system aims to improve the ability of students to understand the problem requirements and verify answers without giving students access to code.

Additional Key Words and Phrases: AI, Pedagogy

ACM Reference Format:

David Cooper. 2026. Coding Assignment Demo System. In *Proceedings of CS5704*. ACM, New York, NY, USA, 3 pages. <https://doi.org/XXXXXX.XXXXXX>

1 Introduction

Almost all education involves students doing assignments created by the instructor. The primary purpose of these assignments is to enhance learning for the relevant subject areas. In computer science education, one issue that can come up is the assignment is challenging for students to understand. Generally, understanding the assignment is not one of the learning outcomes, so instructors want to ease that process as best they can. An example of an assignment which may be challenging to understand is an assignment which I created about calculating risk probabilities using dynamic programming [1]. This assignment is rather complex, especially for students who are unfamiliar with the game it is based on. Later on, it will be demonstrated how this assignment can be improved.

Two of the issues with coding assignments are verifiability of results and understanding the problem presented. Verifiability of results comes from complex logic in the solution to assignments where the logic is hard to manually trace for large inputs and it causes the student to not be able to verify the results they got are correct. Understanding problems is exactly as it sounds. Some problem statements are challenging to understand because of complexity or novelty. We introduce a system that allows instructors to add interactable components to their assignments to help deal with these issues.

2 Related Work

The issue presented above is a problem that all instructors have faced and because of that there are many solutions which try to resolve it. The simplest solution which is used almost universally is using examples. By having the student see example inputs and outputs, this helps them verify their solutions as well as enhances understanding of the problem statement. However, examples are static and may not cover all use cases or be sufficient for the student to grasp the complexities of the problem.

Author's Contact Information: David Cooper, davidcooper@vt.edu, Virigina Tech, Blacksburg, Virginia, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

Another class of solutions which attempts to mitigate issues with comprehension is algorithm visualization. Algorithm visualization can help students see the inner workings of different algorithms and has been shown, although with mixed results, to enhance student learning [3]. For the purposes of assignments, algorithm visualization has some drawbacks. First, if the development of the algorithm is part of the assignment, then showing the student the algorithm defeats the purpose of the assignment. Second, algorithm visualizations are often accompanied by code, which also defeats the purpose of assignments.

There are also interactive textbooks which help students understand issues using interactive components. One such example is Virginia Tech’s OpenDSA [2]. However, these interactive textbooks run into the same issues that algorithm visualizations do in regards to having code and not allowing students to develop algorithms themselves.

3 Requirements Elicitation

As part of creating the system to enhance student understanding of assignments, a survey was performed to look at how students and educators would use a system which allows for interactive assignment components. The responses tended to view the system skeptically or did not have a good view of the idea behind it. Two of the responses mentioned viewing code. Because the system is built to work with assignments, giving students access to code would be against the spirit of education. One response mentioned that demoing solutions to students could lead to design fixation and having the student’s response be too close to the demo, although this respondent did mention that the system would seem to be useful for enhancing learning.

Analyzing these results, we can see that the idea of the system is not something which is obvious to students. Unfortunately, this likely means that such a system is not something which students generally believe would be helpful as it is not something they think about. Because of the effort of creating the system and the lack of benefit combined with concerns of academic integrity and design fixation, it should be the case the project would end here. However, this is in itself a class assignment, so production had to continue.

4 Usability Research

To test usability, a sample assignment was created without functionality for students to test. This was created by converting the assignment mentioned earlier [1] into HTML and adding forms and javascript components to allow for base functionality. The responses were split as to the efficacy of the interactive components. One respondent mentioned the interactive components “may not be a good design choice” which raises the question about the usefulness of the project as a whole. Another respondent said “The interactive inputs are intuitive” so the opinion was not shared by all.

Using this feedback, nothing was really changed in the final product. The negativity from this stage of design testing lends to the idea that the project should not commence as well.

5 AI Usage

To develop this system, Claude Code was used. This was my first experience using AI tools for developing software and the experience was pretty great. AI was used because it is something which is used heavily in modern software development and knowledge of it is necessary to be a good instructor in the future. It was used as the primary form of developing the software for the project, writing basically all the code. Some manual changes were made if they seemed easier to implement than explaining to an AI agent.

The overall experience was very positive. It greatly reduced the development time for the project and most features worked when it was first written. However, there were two concerns related to AI usage. First, this project was quite

small. Overall, there are fewer than 1000 lines of code. Whether or not it can scale effectively to larger projects is unknown and how it deals with larger established codebases is unknown. Additionally, setting up Claude Code ran into an issue with VSCode integration that took almost as long to resolve as using the AI tool to develop the system.

6 Implementation

Because the project was determined to be not worth pursuing, implementation was done in a way which prioritized efficiency over long-term stability. The development was very akin to vibe coding, where the AI was asked to develop a section and then tests were performed to verify that it works as intended. As such, the code works however the AI felt was best to create what was asked of it. The result is that the code is a web based application with a Node server. However, there are not even any classes involved in the final code, so discussions about architecture don't make much sense. This is something which works for the small scope of this project but is not a good idea for long term stability or maintenance.

7 Discussion

The issue of increasing assignment understanding is an open question, but the solution discussed in this paper is one that is likely not worth pursuing. Improvements to the created system do exist, such as integrating with Canvas for a more seamless student experience. However, the fundamental issue of not being particularly useful remains. As such, I do not recommend pursuing development of the ideas presented in this paper unless a different novel approach is used.

8 Conclusions

Overall, the project did not go particularly well. An attempt was made to tackle the issues related to students understanding assignments, but the result fell short of a good product. However, there were good learning opportunities to be had. AI development was shown to be very effective for projects of this size. Most importantly, this project serves as a good case study for why software development processes exist. By identifying user needs early, it could have been identified that the approach was not effective. By doing so, the project could have never proceeded and a large amount of development time could have been saved or alternative approaches could have been explored.

References

- [1] [n. d.]. Example Assignment - Portfolio — docs.google.com. <https://docs.google.com/document/d/1u{A}{D}{Z}{b}{R}{v}{D}{b5{X}5ry{F}q{E}m{Y}exv2qb88c4qyvsnk{P}nd{J}{D}j{D}s/edit?usp=sharing>. [Accessed 07-03-2026].
- [2] [n. d.]. OpenDSA — opensa-server.cs.vt.edu. <https://opensa-server.cs.vt.edu/>. [Accessed 07-03-2026].
- [3] Christopher D Hundhausen, Sarah A Douglas, and John T Stasko. 2002. A meta-study of algorithm visualization effectiveness. *Journal of Visual Languages & Computing* 13, 3 (2002), 259–290.

Received 06 March 2026